

Question 2 Universal Approximation Theorem

Part (a) Formal Statement

Theorem . Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be any continuous, bounded, and non-constant function — for instance, the logistic sigmoid $\sigma(t) = (1 + e^{-t})^{-1}$, or more generally any activation that is not a polynomial. Let $K \subset \mathbb{R}^d$ be a compact set, and write $C(K)$ for the Banach space of continuous real-valued functions on K equipped with the supremum norm $\|\cdot\|_\infty$.

For every target function $f \in C(K)$ and every tolerance $\varepsilon > 0$, there exist a positive integer N , output weights $\alpha_1, \dots, \alpha_N \in \mathbb{R}$, input weight vectors $\mathbf{w}_1, \dots, \mathbf{w}_N \in \mathbb{R}^d$, and bias scalars $b_1, \dots, b_N \in \mathbb{R}$ such that the single-hidden-layer network

$$\hat{f}(\mathbf{x}) = \sum_{i=1}^N \alpha_i \sigma(\mathbf{w}_i^\top \mathbf{x} + b_i) \quad (1)$$

satisfies

$$\|f - \hat{f}\|_\infty = \sup_{\mathbf{x} \in K} |f(\mathbf{x}) - \hat{f}(\mathbf{x})| < \varepsilon. \quad (2)$$

The three key ingredients of the theorem are specified below.

- 1. Class of target functions.** The theorem covers every function that is continuous on the domain K . This is the class $C(K)$; no smoothness (differentiability) is assumed beyond continuity.
- 2. Domain assumption.** The domain K must be *compact* — closed and bounded in $\mathbb{R}^d$. A standard choice is the unit hypercube $K = [0, 1]^d$. Compactness is indispensable: it guarantees that f is uniformly continuous, which is precisely what the $\|\cdot\|_\infty$ error bound requires.
- 3. Notion of approximation.** Approximation is measured in the *uniform* (L^∞) sense as in (2). The error $|f(\mathbf{x}) - \hat{f}(\mathbf{x})|$ must be simultaneously below ε at *every* point of K , not merely on average.

Part (b) Proof Sketch and Key Intuition

A full measure-theoretic proof is not required here; instead we outline the constructive argument that underpins the result.

Stage 1 — Building localised bump functions from sigmoids.

A single sigmoid neuron $\sigma(k(x - c))$ with large gain k behaves like a smooth step centred at c . The *difference* of two such steps,

$$\psi_{a,b}(x) = \sigma(k(x - a)) - \sigma(k(x - b)) \xrightarrow{k \rightarrow \infty} \mathbf{1}_{[a,b]}(x),$$

approximates an indicator function on the interval $[a, b]$. In d dimensions, axis-aligned products of such ridge functions localise to hyperrectangular cells, yielding approximate indicator functions on small regions of K .

Stage 2 — Riemann-sum reconstruction of f .

Partition K into M small cells $\{R_j\}_{j=1}^M$ and pick a representative point $\mathbf{c}_j \in R_j$ in each. Because f is uniformly continuous on the compact set K , the cell diameter can be chosen small enough so that f varies by less than $\varepsilon/2$ within every cell. The linear combination

$$\hat{f}(\mathbf{x}) = \sum_{j=1}^M f(\mathbf{c}_j) \psi_{R_j}(\mathbf{x})$$

is therefore within ε of f everywhere on K , and each ψ_{R_j} is itself realised by a small number of sigmoid units (Stage 1). The total unit count N is thus finite (though potentially large).

Connection to classical approximation theory.

Cybenko’s original proof reaches the same conclusion via functional analysis rather than construction. Using the **Hahn–Banach separation theorem** and the **Riesz–Markov representation theorem**, he shows that if the closed linear span of sigmoid units were a proper subspace of $C(K)$, a nonzero signed Borel measure μ orthogonal to every sigmoid unit would exist. A Fourier/moment argument then forces $\mu = 0$, a contradiction. Hornik’s extension uses the **Stone–Weierstrass theorem** to handle any non-polynomial continuous activation.

Part (c) Two Practical Limitations

Limitation 1 — The theorem gives no bound on network size.

The UAT is an *existence* result: it asserts that a suitable N exists but says nothing about how large it must be. For certain target functions the minimum N that achieves a given ε can grow super-polynomially (or even exponentially) in $1/\varepsilon$ or in the input dimension d .

Why this matters in practice. An arbitrarily wide single-hidden-layer network may be computationally infeasible to train or deploy. Moreover, when N is very large relative to the size of the training set, the network is likely to overfit: it memorises training examples rather than learning the underlying function. The UAT therefore provides no actionable guidance on how to choose architecture width for a real problem.

Limitation 2 — Existence of weights $\neq$ ability to find them.

The theorem guarantees that weight values $\{\alpha_i, \mathbf{w}_i, b_i\}$ achieving ε -accuracy exist somewhere in weight space, but it makes no claim about *optimisation* or *generalisation*:

- **Optimisation gap.** Gradient-based training (SGD, Adam, etc.) can get trapped in local minima or saddle points and may never reach the theoretically optimal weights, especially in wide shallow architectures whose loss landscapes are poorly understood.
- **Generalisation gap.** The approximation guarantee is over the entire compact domain, but training uses only a finite sample. A network that attains small training loss need not generalise to unseen inputs; the UAT provides no bound on test error.

Together, these gaps explain why the UAT alone cannot justify that *training* a sufficiently wide network will succeed in practice.

Part (d) ReLU Activations and the Role of Depth

The classical sigmoid-based UAT does not directly cover the **Rectified Linear Unit** $\sigma(t) = \max(0, t)$, since ReLU is unbounded and piecewise linear rather than sigmoidal. Nevertheless, universality is preserved:

- **Single hidden layer.** Leshno et al. (1993) showed that any non-polynomial continuous activation — including ReLU — suffices for universal approximation in a shallow network, so the single-layer UAT extends to ReLU.
- **Depth compensates for bounded width.** Lu et al. (2017) proved that a deep ReLU network of width as small as $d + 4$ (where d is the input dimension) is a universal approximator provided the depth is allowed to grow. This is qualitatively different from the classical result, which fixes depth at one and lets width grow without bound.
- **Exponential expressivity advantage of depth.** For certain families of functions, a depth- L ReLU network with $O(1)$ neurons per layer can represent functions that would require $O(2^L)$ neurons in a single hidden layer. Intuitively, each additional layer *folds* the input space, doubling the number of linear regions that the network can carve out; a single layer cannot replicate this recursive folding efficiently.
- **Hierarchical feature learning.** Deep architectures align naturally with the compositional structure of real-world data (edges $\rightarrow$ textures $\rightarrow$ objects in vision; phonemes $\rightarrow$ words $\rightarrow$ sentences in speech). Shallow networks lack this inductive bias, which partly explains the empirical superiority of deep networks despite identical universal approximation guarantees in theory.

In summary, for ReLU networks *depth is a strictly more efficient resource than width*: it achieves the same expressive power with exponentially fewer parameters, providing the main theoretical motivation for deep learning architectures.